

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Effect of MK-5684 in Male Participants With Severe Renal Impairment (RI) and With End-stage Renal Disease (ESRD) (MK-5684-010) **Official Title:** An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of MK-5684 in Male Participants With Severe Renal Impairment and With End-stage Renal Disease **Short Title/Acronym:** MK-5684-010 **Protocol Number:** MK-5684-010 (Other ID: CA43216) **NCT Number:** NCT06814132 **Posting ID:** 957490228645037065 **Sponsor/Company:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** MK-5684 (Opevesostat; CYP11A1 inhibitor) and Hormone Replacement Therapy (Prednisone, Fludrocortisone acetate, Cortisone) **Phase of Development:** Phase I **Trial Status:** COMPLETED **Medical Monitor:** Merck Sharp & Dohme Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the pharmacokinetics (PK) and levels of MK-5684 in male participants with severe renal impairment (RI) and end-stage renal disease (ESRD) over time, and compare them to levels in a healthy person's body. **Secondary Objectives:** To compare what happens to MK-5684 after hemodialysis in people with severe RI and ESRD, and to evaluate the safety and tolerability of the intervention.

2.2 Background & Rationale MK-5684 (Opevesostat) is an orally active, selective CYP11A1 inhibitor being investigated for the treatment of metastatic castration-resistant prostate cancer (mCRPC). Because the drug suppresses the production of all steroid hormones and precursors, it requires co-administration with hormone replacement therapy (HRT). This study aims to characterize how severe renal impairment and hemodialysis alter the drug's clearance and systemic exposure, ensuring proper and safe dosing regimens for patients with compromised kidney function.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Clinical Pharmacology) **Control Type:** Healthy matched participant cohort **Blinding:** Open-label **Randomization:** N/A (Assigned by cohort based on renal function status)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 24 male participants **Number of Sites:** Multicenter **Study Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria All Participants: 1. Male participants matching age parameters outlined in the protocol. 2. Continuous non-smoker or moderate smoker (≤ 10 cigarettes per day or equivalent) for at least 3 months prior to the first dosing. 3. Body mass index (BMI) ≥ 18.0 and ≤ 42.0 kg/m² at the screening visit. 4. Ability to comply with study protocols, including a 10-hour overnight fast.

Cohort Specific: * **Group 1 (Severe RI):** Has severely impaired renal function as determined by estimated glomerular filtration rate (eGFR) using the 2021 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) creatinine equation. * **Group 2 (ESRD):** Has ESRD maintained on stable outpatient regimen of intermittent high-flux hemodialysis (HD) (at least 3 times per week) for a minimum of 3 hours per dialysis session, using a complication-free well-maintained arteriovenous fistula or arteriovenous graft, for at least 3 months prior to first dosing and is expected and planning to continue HD. * **Group 3 (Healthy Control):** Has normal renal function.

4.2 Exclusion Criteria All Participants: 1. Female sex. 2. History or presence of any of the following: Adrenal insufficiency; Hepatic impairment; Clinically significant hypotension; cardiac arrhythmia; cardiac conduction abnormalities or recurrent unexplained syncopal events; Second- or third-degree atrioventricular (AV) heart block (except in participants with a functional artificial pacemaker); Clinically significant sick sinus syndrome; Any systemic fungal infection; Hypothyroidism; Chronic infection. 3. Has a first-degree relative with multiple unexplained syncopal events, unexplained cardiac arrest, or sudden cardiac death, or has a known family history of an inherited arrhythmia syndrome (including Brugada syndrome). 4. History of cancer (malignancy). 5. Positive for human immunodeficiency virus (HIV), hepatitis B surface antigen (HBsAg), or hepatitis C virus (HCV). 6. Intolerance to the required Hormone Replacement Therapy components (Prednisone, Fludrocortisone acetate, Cortisone) or MK-5684 components.

Cohort Specific: * **Group 1 (Severe RI):** History or presence of renal artery stenosis; had a renal transplant; has ESRD requiring HD. * **Group 2 (ESRD):** Required frequent emergent HD (≥ 3) within a year prior to the first dosing; history or presence of renal artery stenosis; had a functioning renal transplant within the past 2 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single oral dose of MK-5684 tablet evaluated across two periods (Period 1 and Period 2). Hormone replacement therapy (HRT) including cortisone derivatives is administered approximately 4.5 hours after the MK-5684 dose. **Route of Administration:** Oral (Tablets for MK-5684, Prednisone, Fludrocortisone acetate, Cortisone). **Duration of Treatment:** Single doses per period with specified washouts, encompassing intensive pharmacokinetic sampling.

5.2 Concomitant & Prohibited Therapy Permitted: Routine hemodialysis (HD) and standard background medications required for ESRD. **Prohibited:** Strong CYP inhibitors or inducers that could interfere with the CYP11A1 mechanism or the PK readings.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Renal Function & eGFR Labs
Period 1	Day 1	10-hour overnight fast; single oral MK-5684 dose 30 min prior to scheduled HD; HRT given under fed conditions 4.5h post MK-5684; intensive PK sampling.
Period 2	Follow-up	10-hour overnight fast; single oral MK-5684 dose immediately followed by scheduled HD; HRT given under fed conditions 4.5h post MK-5684; intensive PK sampling.
End of Study	Follow-up	Final Safety Labs, AE review, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- AUC_{0-inf} of MK-5684:** Area Under the Concentration Versus Time Curve from 0 to Infinity after a single dose of MK-5684 (determined via plasma samples).
- AUC_{0-last} of MK-5684:** Area Under the Concentration Versus Time Curve From 0 to the Time of the Last Quantifiable Sample (determined via plasma samples).
- AUC₀₋₂₄ of MK-5684:** Area Under the Concentration Versus Time Curve From 0 to 24 Hours After Dosing (determined via plasma samples).

7.2 Secondary Outcomes

- Number of Participants who Experience an Adverse Event (AE).
- Number of Participants who Discontinue Study Intervention Due to an AE.
- Dialysis Clearance (CLD) of MK-5684 Based on Plasma.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Onsite monitoring during confinement/dosing periods for clinical signs, vital signs, and standard laboratory abnormalities. **Serious Adverse Events (SAEs):**

Must be reported to the Sponsor/pharmacovigilance team within 24 hours. **Data Monitoring Committee (DMC):** Internal safety review overseen by Merck Sharp & Dohme LLC.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK Parameter Set (all treated patients with sufficient plasma data) and Safety Set (all patients receiving ≥ 1 dose of MK-5684). **Sample Size Justification:** Target Enrollment = 24 configured to provide standard statistical power to define the PK differences (Geometric Mean Ratios) between renally impaired groups and healthy volunteers. **Handling of Missing Data:** Plasma concentrations below the Limit of Quantification (BLQ) handled as zero pre-dose, per standard Phase 1 PK modeling guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Onsite Phase 1 clinical monitoring and bioanalytical site auditing to ensure strict dosing and sampling time compliance. **Audit Trail:** Full adherence tracking for fasting states, eCRF locking, and sample chain of custody.

11. Study Identifiers & Governance

Primary ID: MK-5684-010 **Registry Number:** NCT06814132 **Posting ID:** 957490228645037065 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** A Study to Evaluate the Effect of MK-5684 in Male Participants With Severe Renal Impairment (RI) and With End-stage Renal Disease (ESRD) (MK-5684-010) **Official Regulatory Title:** An Open-Label, Single-Dose Study to Evaluate the Pharmacokinetics of MK-5684 in Male Participants With Severe Renal Impairment and With End-stage Renal Disease

12. Clinical Framework (Renal Impairment Specific)

Disease Areas: Kidney Failure, Chronic; End-Stage Kidney Disease; Renal Failure, Chronic; Renal Failure, End-Stage; Healthy Participants. **Preferred UMLS Name:** End-Stage Renal Disease (MeSH: D007676) **Medical Methodology:** Pharmacokinetic (PK) tracking coupled with necessary Hormone Replacement Therapy (HRT) **Optimization Target:** PK modeling to instruct specific dosing labels for renal failure patients

13. Study Procedures & Methodology

Management Pathway: Controlled clinical pharmacology setting coordinating with hemodialysis scheduling **Education Protocol (HCP):** Strict adherence training to the exact timing of MK-5684 dosing, subsequent hemodialysis, and HRT administration **Education Protocol (Patient):** Compliance training on the 10-hour overnight fasting conditions and standard meal rules post-dose **Quality Audit Mechanism:** Phase 1 bioanalytical tracking and real-time AE observation

14. Observation & Follow-Up Schedule

Total Duration: Limited-duration tracking spanning the primary PK profiling periods. **Onsite Visit Cadence:** Clinic confinement required during Period 1 and Period 2 for intensive serial blood draws. **Remote Monitoring Frequency:** Standard Phase 1 phone follow-up post-clinic discharge. **Checklist Review Interval:** Continuous clinical observation during dosing days.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				